

```
// SMART BLIND STICK USING ARDUINO UNO
```

```
#define TRIG 9  
#define ECHO 10  
#define WATER A0  
#define BUZZER 8
```

```
long duration;  
int distance;  
int waterValue;
```

```
const int obstacleDistance = 50; // Detect obstacle within 50 cm  
const int waterThreshold = 800; // Rain sensor threshold
```

```
void setup()  
{  
  pinMode(TRIG, OUTPUT);  
  pinMode(ECHO, INPUT);  
  pinMode(BUZZER, OUTPUT);
```

```
  digitalWrite(BUZZER, LOW);
```

```
  Serial.begin(9600);  
}
```

```
void loop()  
{  
  // Measure Distance  
  digitalWrite(TRIG, LOW);  
  delayMicroseconds(2);
```

```
  digitalWrite(TRIG, HIGH);  
  delayMicroseconds(10);
```

```
  digitalWrite(TRIG, LOW);
```

```
  duration = pulseIn(ECHO, HIGH, 30000);
```

```
  if (duration == 0)  
    distance = 999;  
  else  
    distance = duration * 0.034 / 2;
```

```
  // Read Water Sensor
```

```
waterValue = analogRead(WATER);
```

```
Serial.print("Distance: ");
```

```
Serial.print(distance);
```

```
Serial.print(" cm ");
```

```
Serial.print("Water: ");
```

```
Serial.println(waterValue);
```

```
// Buzzer OFF by default
```

```
digitalWrite(BUZZER, LOW);
```

```
// Obstacle Detection
```

```
if (distance > 0 && distance <= obstacleDistance)
```

```
{
```

```
digitalWrite(BUZZER, HIGH);
```

```
}
```

```
// Water Detection
```

```
else if (waterValue < waterThreshold)
```

```
{
```

```
digitalWrite(BUZZER, HIGH);
```

```
}
```

```
else
```

```
{
```

```
digitalWrite(BUZZER, LOW);
```

```
}
```

```
delay(100);
```

```
}. In this what is the data base used and web frame work used and also wbt the frontend stack  
in single line
```